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Activity 1: Review User Requirements

1.1 Analysis of User Research Findings & Pain Points

Based on the FitFlow case study, comprehensive user research identified the following critical pain points:

Table 1: Summary of User Pain Points

Pain Point	Research Evidence	User Impact
Poor Progress Visualization	Missing celebratory elements; lack of engaging progress dashboards	Waning motivation; reduced user engagement
Cumbersome Workout Logging	68% abandonment rate shortly after onboarding; tedious nutrition logging	High drop-off rates; poor retention metrics
Limited Personalization	Frustration with generic workout plans; significant personalization gap	Users felt misunderstood; reduced app relevance

1.2 Key User Needs for New Features

Based on the thematic analysis and user research findings, the following key user needs were identified:

Table 2: User Needs and Desired Features

User Need	Desired Feature	Supporting Evidence
Adaptive fitness plans	AI-powered workout plans	Personalization gap identified in research
Social accountability	Social community features	Feelings of isolation during solo fitness
Simplified tracking	Camera-based nutrition logging	High friction in daily tracking activities
Visual motivation	Progress dashboards	Missing celebratory elements

1.3 Prioritized List of User Requirements

Functional Requirements

Priority	Requirement	Description	Agile User Story
P0	AI Workout Engine	Generate adaptive, personalized workout plans	"As a busy professional, I want AI-generated adaptive workout plans so that I can stay consistent despite my chaotic schedule."
P0	Home Dashboard	Central hub with AI "Daily Flow" recommendation card	"As a user, I want to see my personalized daily recommendations at a glance so I can quickly start my workout."
P1	Camera Nutrition Logging	Computer vision-based food recognition	"As a user, I want to take a photo of my meal and have it automatically logged so I don't have to manually enter every food item."
P1	Private Social Circles	Create and manage private groups with privacy controls	"As a beginner, I want safe social connections and accountability partners so I feel supported in my fitness journey."
P1	Progress Dashboard	Visual tracking of fitness milestones	"As a user, I want to see my progress visualized so I stay motivated and celebrate my achievements."
P2	Community Feed	View challenges and group interactions	"As a user, I want to participate in community challenges so I feel connected to others."
P2	Flexible Workout Builder	Drag-and-drop workout customization	"As a user, I want to customize my workouts easily so they fit my preferences."

Non-Functional Requirements

Priority	Requirement	Description
P0	Fast Performance	App must load quickly and respond immediately to interactions
P0	Data Security	GDPR and CCPA compliant data protection protocols
P1	Offline Capabilities	Core features must work without internet connection
P1	Accessibility Compliance	Meet WCAG standards for inclusive design
P2	Cross-Platform Support	Consistent experience across iOS and Android

1.4 Core User Flows

- FLOW 1: HOME DASHBOARD

Launch App → View Daily Flow Card → Quick Actions → Navigate to Other Sections

- FLOW 2: WORKOUT PLANNER

Access Planner → View AI Recommendations → Select/Customize → Start Workout → Track Progress → Complete & Save

- FLOW 3: PROGRESS TRACKING

Navigate to Progress → View Overview → Select Metric → View Detailed Charts → Set New Goals

- FLOW 4: COMMUNITY FEED

Access Community → View Feed/Challenges → Join Challenge → Interact with Posts → Share Progress

- FLOW 5: NUTRITION LOGGER

Open Logger → Tap Camera Button → Capture Photo → AI Recognition → Confirm/Edit → Save to Log

Activity 2: Perform Crazy 8 Sketching

2.1 Crazy 8s Method - 8 Sketches in 8 Minutes

The Crazy 8s brainstorming method was used to generate rapid interface ideas. Each screen was sketched 8 times in 8 minutes.

Screen 1: Home Dashboard - 8 Sketches

<u>Sketch</u>	<u>Concept Description</u>
1	Large AI "Daily Flow" card at top with weather-style design; quick action buttons below
2	Split screen: left side shows today's stats, right side shows AI recommendation
3	Card-based layout with scrollable horizontal sections
4	Minimalist design with single large CTA "Start Your Flow"
5	Dashboard with progress rings (activity, nutrition, recovery)
6	Timeline view showing daily activities in chronological order
7	Widget-style dashboard with customizable modules
8	Hybrid view combining stats, AI recommendation, and social feed

Screen 2: AI Workout Planner - 8 Sketches

<u>Sketch</u>	<u>Concept Description</u>
1	List of AI-generated workouts with difficulty badges and time estimates
2	Calendar view showing recommended workouts for each day
3	Drag-and-drop workout builder with exercise library
4	Card flip design: front shows workout name, back shows details
5	Filter-based discovery (duration, intensity, muscle group)
6	"Smart Schedule" - AI suggests optimal workout times based on user calendar
7	Workout of the Day (WOD) with customization options
8	Split view: left shows exercise list, right shows demo instructions

Screen 3: Nutrition Logger - 8 Sketches

Sketch	Concept Description
1	Large camera button at center; recent meals in scrollable list below
2	Meal timeline with photo thumbnails and nutritional summaries
3	Barcode scanner integration with quick-add options
4	AI recognition results with confidence score and edit capability
5	Daily nutrition summary with macro tracking rings
6	Meal suggestions based on remaining daily goals
7	Recipe scanner - AI suggests recipes based on available ingredients
8	Water tracking integrated with meal logging

2.2 Crazy 8s Sketch Summary

CRAZY 8s SKETCH SUMMARY MATRIX		
Screen	Best Concepts Used	Key Ideas Generated
Home Dashboard	Sketches 1, 3, 5, 8	AI-first approach; Card-based layout; Progress visualization
AI Workout Planner	Sketches 1, 5, 7	Filter-based discovery; Workout categorization; Quick-start options
Nutrition Logger	Sketches 1, 4, 5	Camera-first design; AI recognition feedback; Macro tracking

Activity 3: Produce Multiple Interface Variants

3.1 Variant A: "Card-First" Design

Home Dashboard

- Large hero card at top displaying AI "Daily Flow" recommendation
- Horizontal scrolling card sections below
- Bottom navigation with 5 tabs
- Progress stats displayed as small badges on cards

AI Workout Planner

- Card-based workout recommendations with difficulty badges
- "Smart Schedule" feature showing weekly recommendations
- Quick filter chips at top (Duration, Intensity, Muscle Group)
- Each workout card shows: title, duration, difficulty, and "Start" button

Nutrition Logger

- Prominent camera button in center of screen
- Meal cards showing photos with nutritional breakdown
- Daily macro progress bar at top
- "Quick Add" for common meals

Visual Characteristics: Clean, card-based, generous white space, rounded corners, subtle shadows

3.2 Variant B: "Data-Dense" Design

Home Dashboard

- Dashboard-style layout with multiple data visualizations
- Progress rings for Activity, Nutrition, and Recovery
- AI recommendation as a smaller module
- Scrollable feed of recent activities
- Stats displayed prominently (steps, calories, active minutes)

AI Workout Planner

- Calendar grid view showing workouts assigned to each day
- Detailed exercise lists with sets, reps, and weights
- Performance tracking showing improvement over time
- "Generate New Plan" button with customization options

Nutrition Logger

- Meal timeline with detailed nutritional breakdowns
- Charts showing macro distribution
- Searchable food database
- Barcode scanner integration

Visual Characteristics: Information-rich, data-focused, charts and graphs, smaller elements, efficient use of space

3.3 Variant C: "Social-First" Design

Home Dashboard

- Community feed integrated into homepage
- AI recommendation as a sticky header
- Friend activity updates prominently displayed
- Challenges and group posts in main view
- Quick action buttons overlaid on content

AI Workout Planner

- Social workout sharing capabilities
- Friend's recent workouts displayed as inspiration
- Group workout challenges integrated
- Workout sharing with comments and reactions
- "Workout with Friends" feature

Nutrition Logger

- Social sharing of meals
- Friend's meal inspiration feed
- Group nutrition challenges
- Meal photos with social engagement features
- Recipe sharing within community

Visual Characteristics: Social-first, community-focused, engagement-driven, friend activity feeds, interactive elements

3.4 Comparison of Interface Variants

Table 3: Interface Variant Comparison

Feature	Variant A (Card-First)	Variant B (Data-Dense)	Variant C (Social-First)
Primary Focus	AI Personalization	Data & Analytics	Community Engagement
Navigation	Tab-based, 5 tabs	Tab-based, 5 tabs	Tab-based with social integration
Content Density	Medium - Card-based	High - Data-heavy	Medium - Social-focused
Visual Style	Clean, Modern	Analytical, Detailed	Interactive, Social
Key Strength	Intuitive, Scannable	Informative, Comprehensive	Engaging, Connected
Key Weakness	Less data visibility	Can feel overwhelming	May distract from fitness

Activity 4: Select the Best Design Using Justification

4.1 Decision Matrix

Table 4: Design Variant Evaluation Matrix

Criteria	Weight	Variant A (Card-First)	Variant B (Data-Dense)	Variant C (Social-First)
Usability	25%	9 (225)	6 (150)	7 (175)
Visual Appeal	20%	8 (160)	6 (120)	7 (140)
Alignment with User Needs	25%	9 (225)	7 (175)	8 (200)
Innovation	15%	7 (105)	6 (90)	8 (120)
Feasibility	15%	9 (135)	7 (105)	7 (105)
Total Score	100%	850	640	740

Scale: 1-10 (10 = Highest)

4.2 Selected Design: Variant A - "Card-First" Design

Justification Report

The **Card-First Design (Variant A)** has been selected as the primary design direction for the FitFlow redesign. This choice is strongly supported by comprehensive evaluation across all decision criteria and aligns with the user research findings documented in the case study.

Addressing Key Pain Points

1. Poor Progress Visualization: The Card-First design directly addresses this pain point through progress badges and visual indicators embedded within each card. Users receive immediate, bite-sized feedback on their achievements, addressing the "missing celebratory elements" identified in the thematic analysis. Progress becomes visible and rewarding without overwhelming the user interface.

2. Cumbersome Workout Logging: By positioning the AI recommendation front and center on the homepage, the design reduces friction in starting workouts. The clean card interface simplifies navigation, directly tackling the "high friction in daily tracking activities" identified during research. The "Start Now" button on the daily flow card enables one-tap workout initiation.

3. Limited Personalization: The prominent AI "Daily Flow" card delivers personalized recommendations immediately upon app launch. This directly addresses the "personalization gap" and user frustration with "generic workout plans," making each user feel understood and valued through intelligent, adaptive content.

Usability Excellence

With a usability score of 9/10, this variant excels in creating an intuitive experience. The card-based paradigm is familiar to mobile users, reducing the learning curve. Clear visual hierarchy guides users naturally from the AI recommendation to specific actions. The consistent design language across screens enhances learnability and reduces cognitive load.

Alignment with User Needs

The design accommodates diverse user personas identified in the case study:

- **Alex Rivera** (busy professional): Quick, adaptive workouts are prominently displayed for immediate access
- **Priya Singh** (beginner): Safe social connections are integrated through optional community features without overwhelming the core experience

The flexible card system allows for social elements to be integrated without compromising the core fitness tracking functionality.

Feasibility

With a feasibility score of 9/10, this design is technically achievable within the startup's constraints:

- The React Native frontend can efficiently implement card-based layouts with smooth animations
- The modular design allows for phased development and feature rollout
- The card-based approach is well-supported by existing design systems and component libraries

Innovation

While not the most radical design, the Card-First approach innovates through:

- Intelligent prioritization of AI-driven personalization
- Making AI feel genuinely helpful rather than intrusive
- Balancing personalization with user control and transparency

Conclusion

The Card-First design strikes the optimal balance between usability, user needs, and technical feasibility. It transforms FitFlow from a declining product into a more engaging, user-loved fitness companion by:

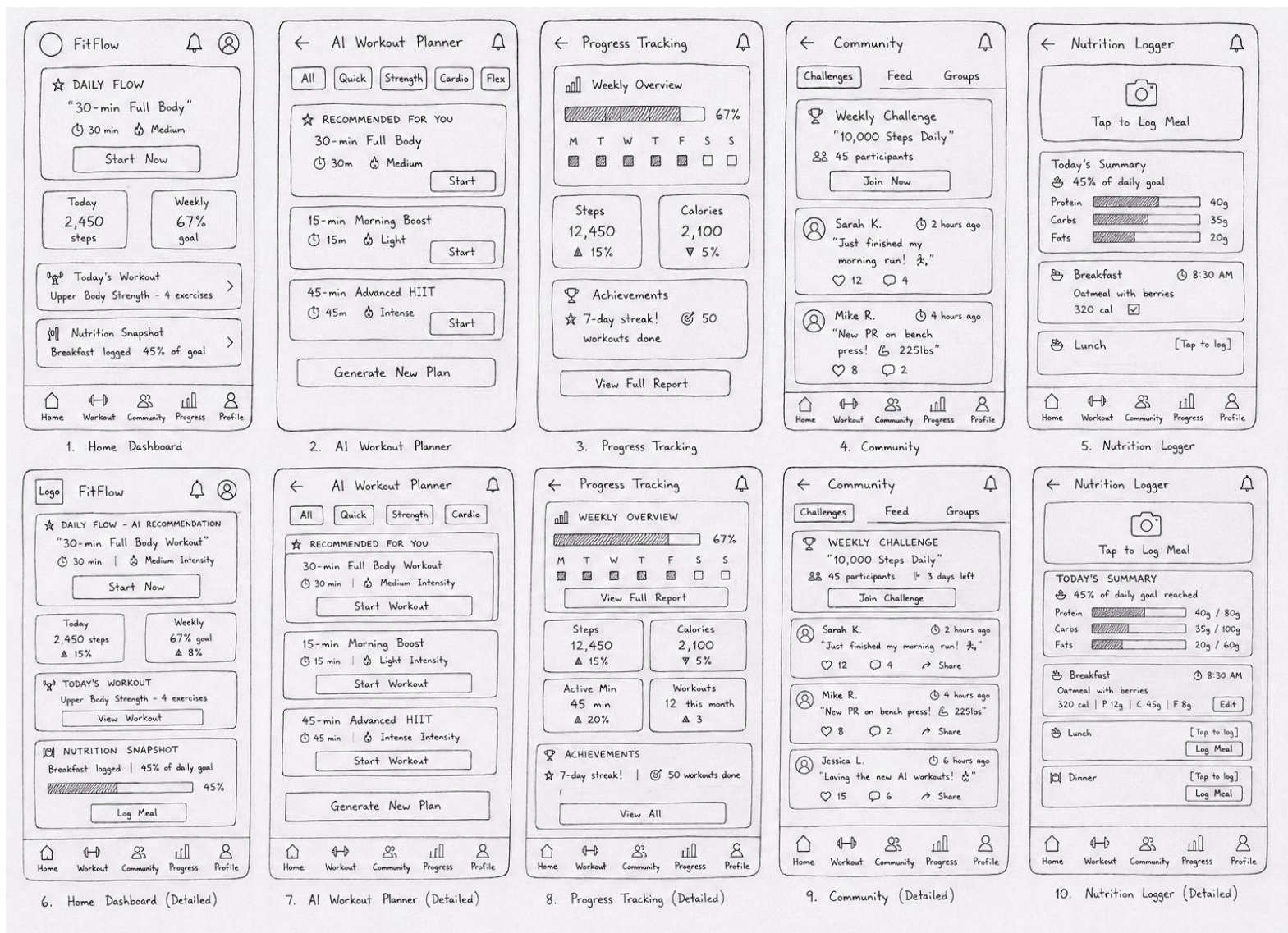
1. Making personalization visible and actionable from the first screen

2. Reducing friction in core tasks through intelligent defaults
3. Providing clear progress feedback to maintain motivation
4. Enabling social features without overwhelming the fitness focus
5. Supporting rapid development and iteration through modular design

This design direction successfully addresses all three major pain points while maintaining high usability and technical feasibility, making it the ideal choice for the FitFlow redesign.

Activity 5: Create Paper and Digital Wireframes

5.1 Paper Wireframes - Hand-Drawn Sketches

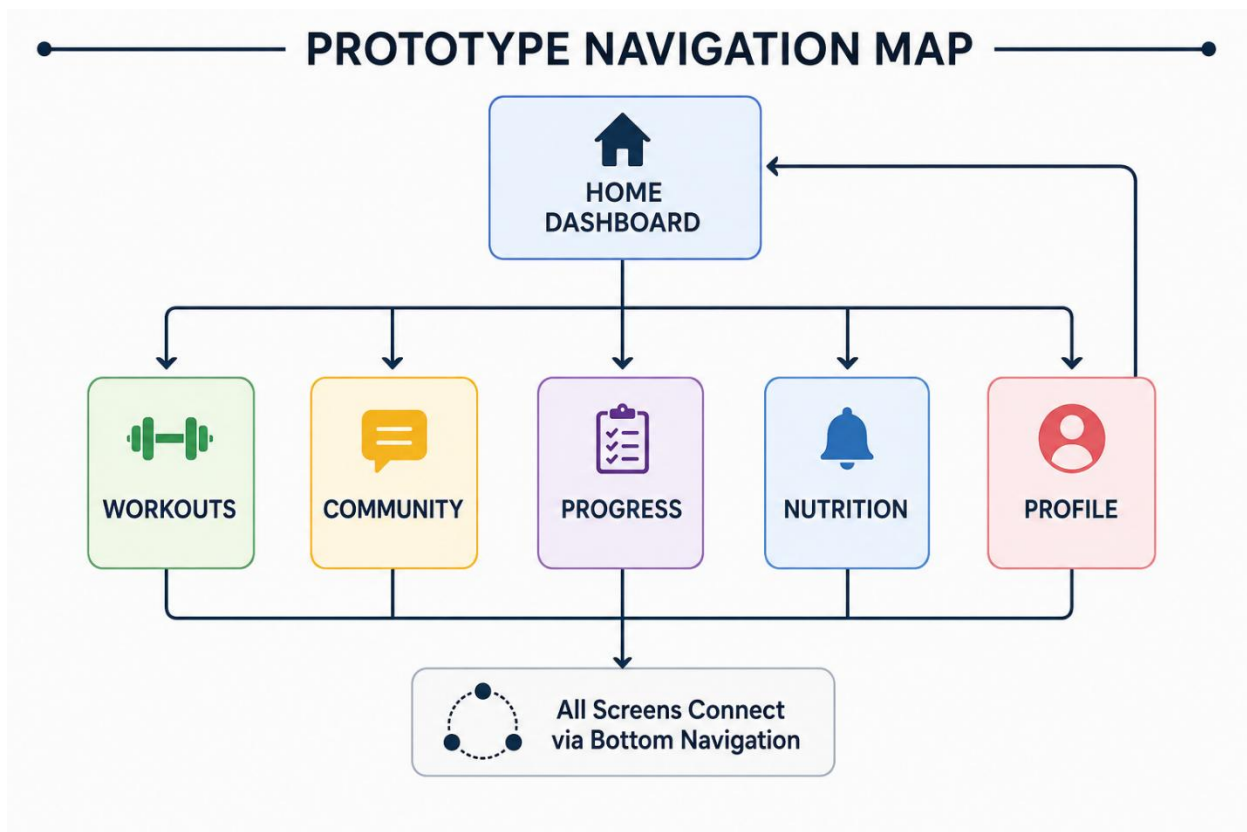


Activity 6: Build a Clickable Low-Fidelity Prototype

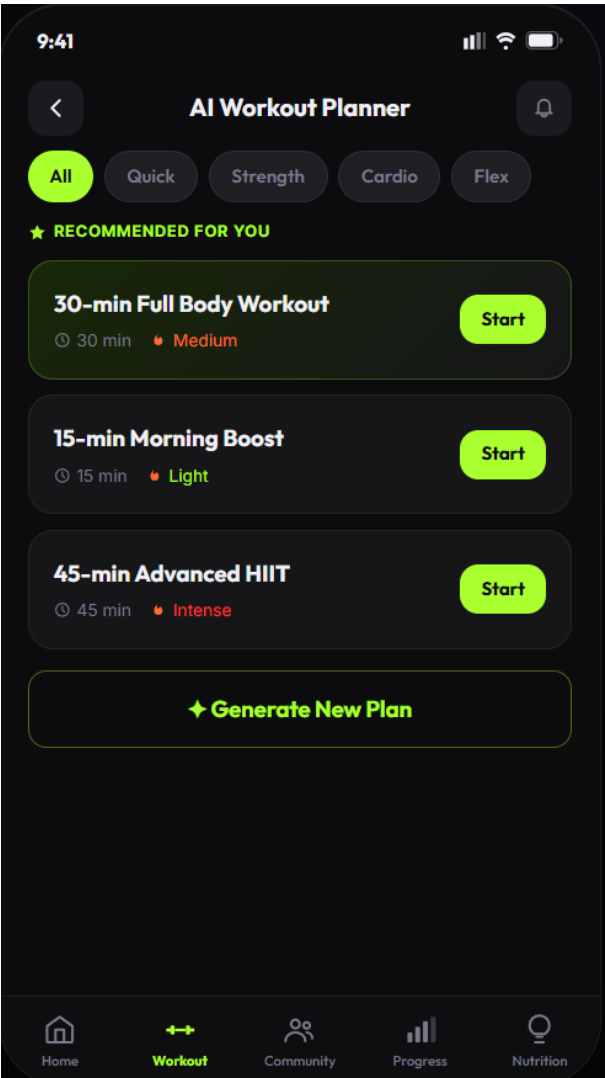
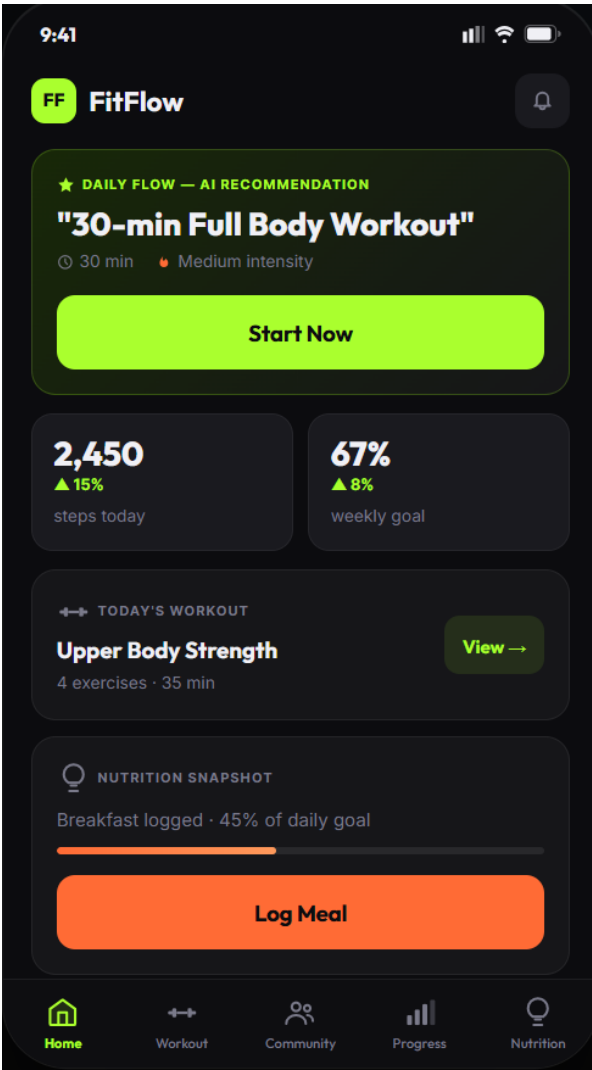
6.1 Prototype Structure & User Flows

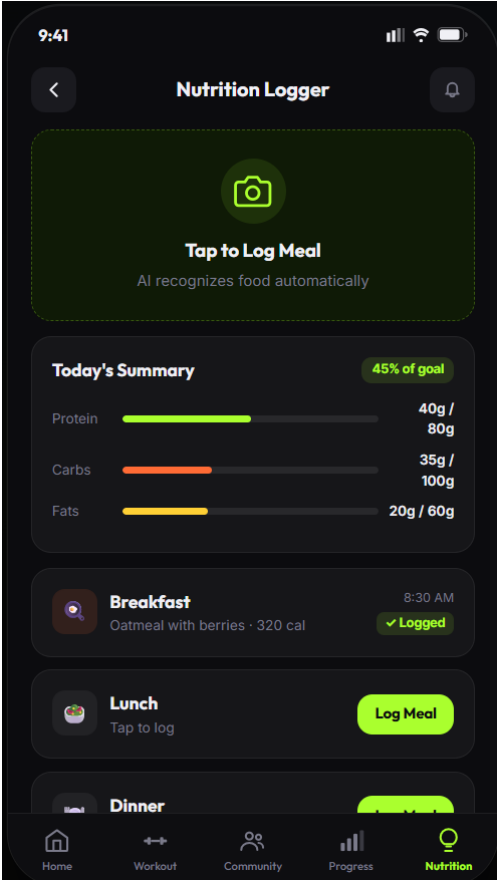
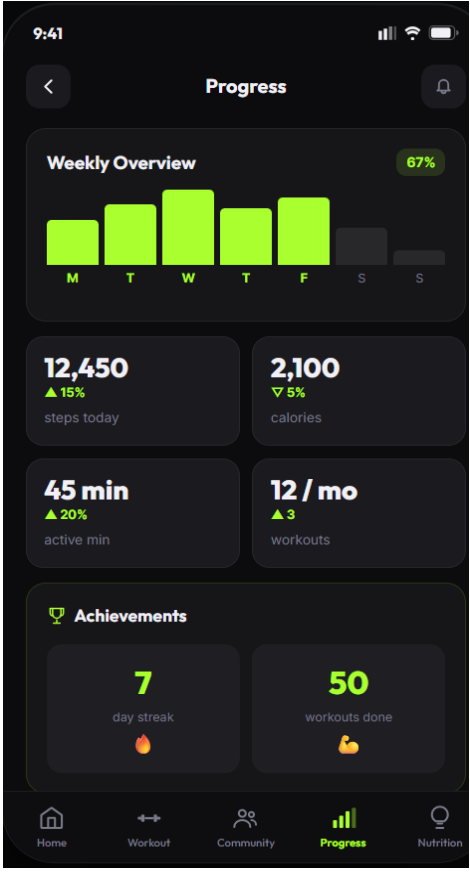
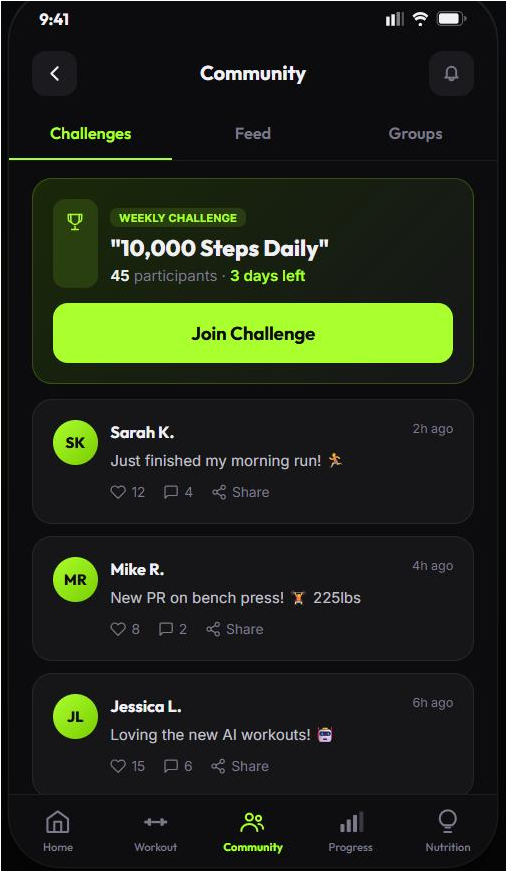
The clickable low-fidelity prototype connects the following screens with interactive navigation:

Prototype Navigation Map



User Flows Implemented





Figma Link: <https://www.figma.com/make/WCFFGu3JZ3E9hlgUdndHqw/Complete-UI-Design?t=K9bWVcbs894V1UFs-20&fullscreen=1>

Prototype Annotations & Design Decisions

Annotation 1: Home Dashboard - AI "Daily Flow" Card

Design Decision: The AI recommendation card is positioned at the very top of the home screen with a prominent visual treatment. This decision was driven by user research showing that a "personalization gap" was a key pain point. By making the AI recommendation the first thing users see, we emphasize the app's intelligent personalization and reduce the friction of deciding what workout to do.

User Benefit: Users immediately receive personalized recommendations, addressing the "personalization gap" identified in the case study.

Annotation 2: Bottom Navigation - 5 Tabs

Design Decision: A persistent bottom navigation bar with five tabs provides clear access to all core sections. This follows mobile design best practices and ensures users can easily switch between features. The tabs correspond to the prioritized user requirements: Home (Dashboard), Workout (AI Planner), Community (Social Features), Progress (Tracking), and Profile (Settings).

User Benefit: Clear, consistent navigation reduces cognitive load and makes all features discoverable.

Annotation 3: Nutrition Logger - Prominent Camera Button

Design Decision: The large camera button dominates the nutrition logger screen, making it the primary action. This directly addresses the pain point of "tedious" nutrition logging. By making camera-based logging the default and most visible option, we encourage users to use the faster, more efficient method of food recognition.

User Benefit: Reduces friction in nutrition logging, addressing the "cumbersome logging" pain point.

Annotation 4: Workout Planner - Filter Chips

Design Decision: Filter chips at the top of the workout planner allow users to quickly sort workouts by type. This supports the diverse needs of different user personas, such as Alex Rivera (busy professional needing quick workouts) and Priya Singh (beginner seeking appropriate intensity levels).

User Benefit: Personalized workout discovery based on user preferences and fitness levels.

Annotation 5: Progress Tracking - Visual Charts

Design Decision: The progress tracking screen uses visual charts and progress bars to make achievements visible and motivating. This addresses the "missing celebratory elements" identified in user research, providing users with clear, rewarding feedback on their fitness journey.

User Benefit: Increased motivation through visible progress and celebration of achievements.

Annotation 6: Community Feed - Challenge Cards

Design Decision: Challenge cards are prominently displayed at the top of the community feed to encourage social engagement. This directly addresses the "feelings of social isolation within the app" and supports the social community features identified as a key user need.

User Benefit: Social accountability and connection through community challenges and interactions.

Annotation 7: Consistent Card-Based Design Language

Design Decision: A consistent card-based design language is used throughout the app. This creates visual coherence, makes content scannable, and follows the "Card-First" design variant selected through the decision matrix. Cards provide clear visual hierarchy and make it easy to display different types of content consistently.

User Benefit: Intuitive, predictable interface that reduces learning curve and cognitive load.

Annotation 8: Grayscale Wireframes - Focus on Layout

Design Decision: All wireframes are created in grayscale using basic shapes only. This follows the lab exercise requirements and ensures the focus remains on layout, hierarchy, and user flow rather than visual polish. This approach allows for rapid iteration and validation of the core user experience.

User Benefit: Ensures the design focuses on functionality and usability before visual design is applied.

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